4.5.5 Wrap-Up: Error Analysis

The key features and the interpretations given for the context have errors. Find one of the errors and fix it.

1. The average price of a gallon of orange juice from October 2013 to September 2014 can be modeled by the function f shown, where x represents the number of months since October 2013.

$$f(x) = 6.12 + 0.03x$$

	Domain	Range	x-intercept
Key Feature	$x \geq 0$	$y \geq 6.12$	6.12

Interpretations	
Slope	**y-intercept**
For every gallon of orange juice, the month increased by 0.03.	In October 2013, the average price of a gallon of orange juice was $\$6.12$.

Unit 4, Lesson 6: Determining Constraints

Benchmarks

MA.912.AR.2.5 Solve and graph mathematical and real-world problems that are modeled with linear functions. Interpret key features and determine constraints in terms of the context.

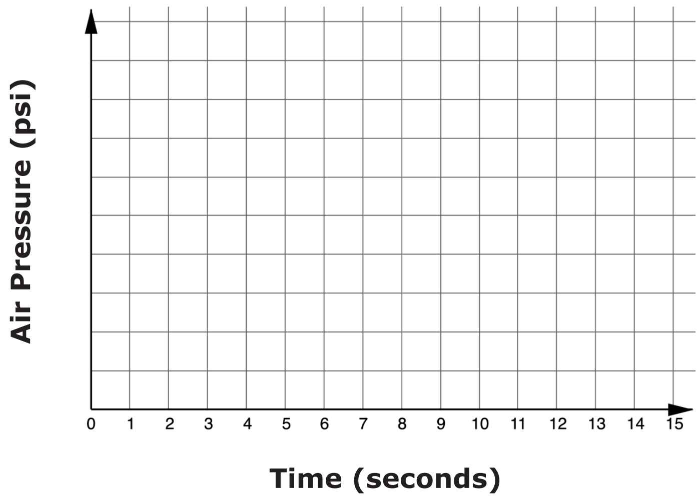
MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.

Learning Targets

- ☐ I can graph a line and interpret key features.
- ☐ I can determine constraints in terms of the context.
- ☐ I can evaluate and interpret function notation in terms of context.

4.6.1 Warm-Up: Air Pressure

1. As you watch the video of a football being pumped up, use the graph to plot the pressure in pounds per square inch (psi) over time.



2. How does your graph compare to the graph at the end of the video?

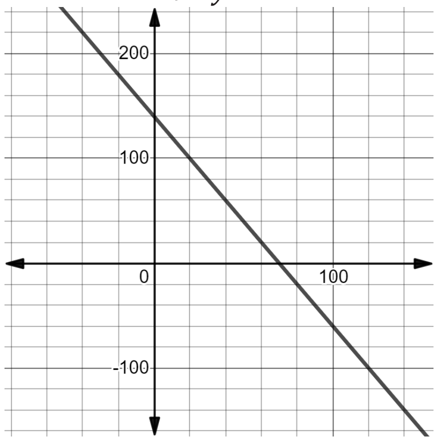
3. The person pumping the football paused on occasion. What would the graph look like if the person did not pause?

4. Explain why it is possible that the y-intercept is greater than 0.

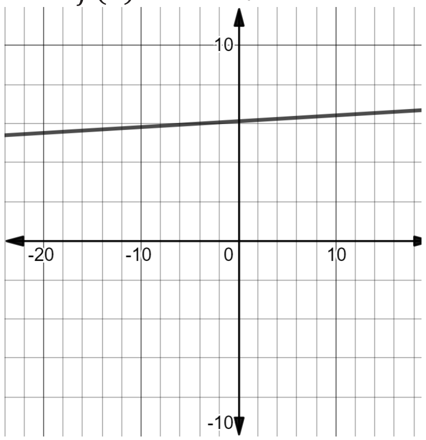
4.6.2 Guided Instruction: Constraints

1. Find the domain and range for each line.

$16x + 8y = 1120$



$f(x) = 6.12 + 0.03x$



Domain	Range

Domain	Range

2. A domain and a range are given for each context.

Context	
Lake Wales Little Theater is an intimate 140-seat theater. The theater uses the equation $16x + 8y = 1120$ to ensure each performance earns $\$1{,}120$ to cover the costs of a children's production. The variable x represents the number of adult tickets and y represents the number of child tickets.	
Domain	**Range**
$\{x: x = 0, 1, 2, \ldots 70\}$	$\{y: y = 0, 1, 2, \ldots 140\}$

Context	
The average price of a gallon of orange juice from October 2013 to September 2014 can be modeled by the function $f(x) = 6.12 + 0.03x$, where x represents the number of months since October 2013.	
Domain	**Range**
$\{x: 0 \leq x \leq 12\}$	$\{y: y \geq 6.12\}$

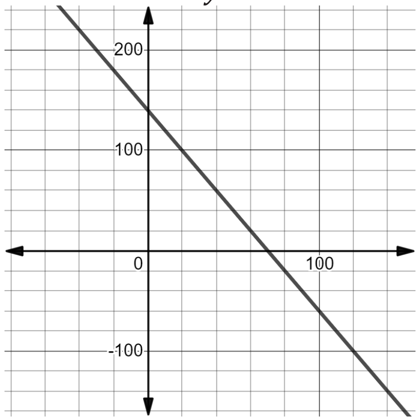
a. Discuss with your partner why the domain for the theater context is not $\{x: x \geq 0\}$.

b. Discuss with your partner why the domain for the orange juice context does not have to be restricted to whole numbers.

The graphs for the two contexts look the same as the graphs without the context. Yet, the context places a restriction on how much the mathematics can represent a situation.

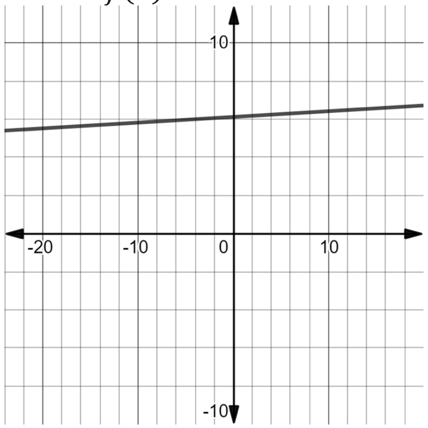
Lake Wales Little Theater

$16x + 8y = 1120$



Average Price of Orange Juice

$f(x) = 6.12 + 0.03x$



A **constraint** is a limitation that is placed on a variable, key feature, or solution so that the mathematical model is realistic.

4.6.3 Collaboration: Constraints

1. A boat rental company has boats available that have a maximum capacity of 8 people and a weight limit of 850 pounds. The company uses an approximate weight of 180 pounds for each adult and 80 pounds for each child.

   a. Discuss with your partner the constraints found in the context.

   b. Select all the inequalities that model constraints for this situation, where x represents the number of adults and y represents the number of children.

   - ☐ $x \geq 0$
   - ☐ $y \geq 0$
   - ☐ $x \leq 180$
   - ☐ $x + y \leq 8$
   - ☐ $x + y \leq 850$
   - ☐ $180x + 80y \leq 850$

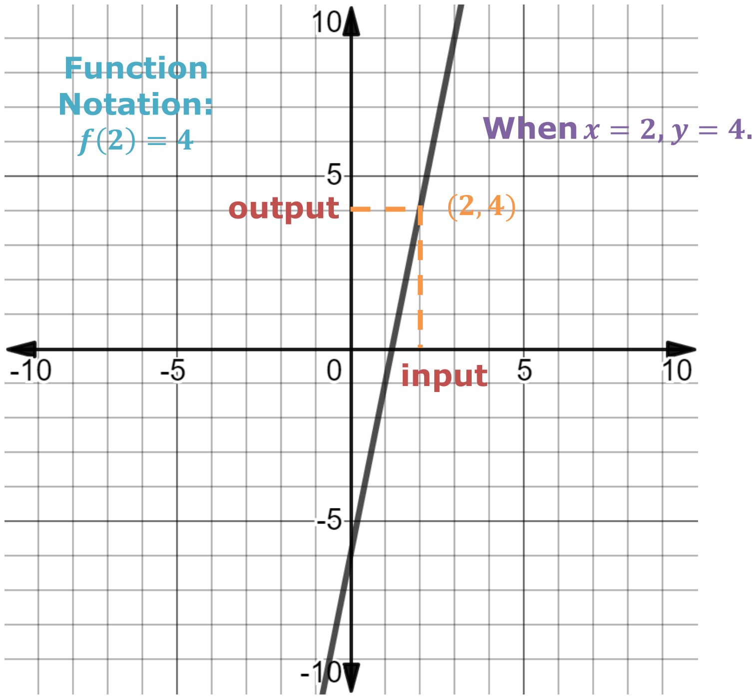
   c. Write the constraints using set-builder notation.

4.6.4 Guided Instruction: Finding Solutions on the Graph

1. Consider the function $f(x) = 5x - 6$.

   a. For $f(x) = 5x - 6$, find $f(7)$.

The function $f(x) = 5x - 6$ is shown on the graph. The value of $f(2) = 4$ is shown.



The graph of a function can be used to find values or solutions. The graph shows the value, $f(2)$. The graph also shows the solution to $f(x) = 4$, which can also be written as the solution to $4 = 5x - 6$.

   b. Solve $-1 = 5x - 6$.

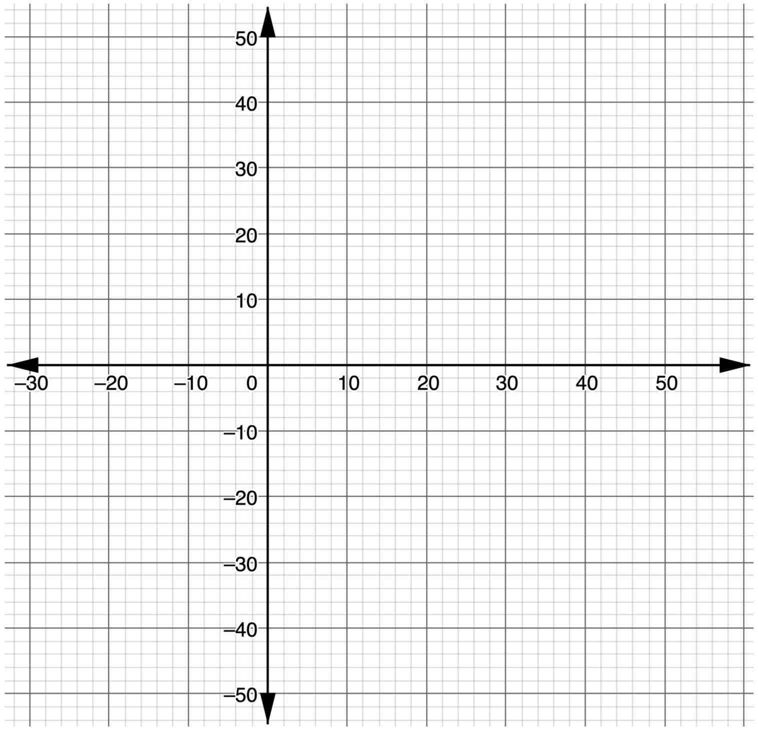
   c. Find the x-value for which $f(x) = -1$.

   d. Find $f(0)$.

   e. What key feature is $f(0)$?

2. The function $S(w) = -0.5(w - 6) + 21$ models the percentage of adults in the U.S. who smoked cigarettes w years after 1997.

a. Graph the function.



b. Find the value of w for which $S(w) = 24$.

c. In terms of the context, what does the value of w for which $S(w) = 24$ mean?

d. What is the name of the key feature found by solving $S(w) = 24$?

e. Rewrite $S(w) = (-0.5)(w - 6) + 21$ in slope-intercept form.

f. Find the solution to $S(w) = 0$.

g. Interpret the solution found for $S(w) = 0$ in terms of the context.

h. What key feature is found when solving $S(w) = 0$?

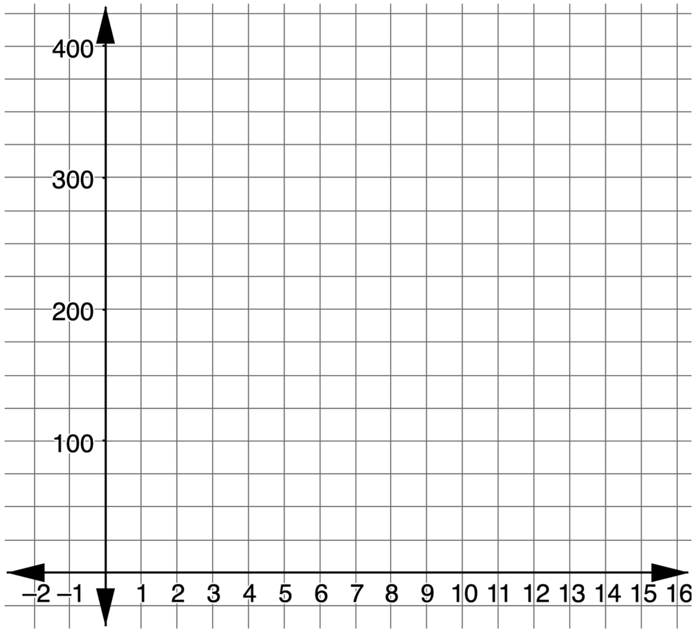
i. Use the graph to determine the year that the percentage of smokers will be 14%.

4.6.5 Try It: Graphing a Line

1. Emmanuel's birthday party will be at a trampoline park. The cost for each person is $\$22$. The cost of the room for the party is $\$75$. Emmanuel's parents have set a budget of $\$400$.

   The function $C(x) = 75 + 22x$ models the total cost, C, where x is the number of people at the party.

   a. Graph the function in terms of the context.

   b. Determine the maximum number of people who can be at the party for the budget of $\$400$.

4.6.6 Wrap-Up: Growth Mindset

The power of "YET."

"The power of believing that you can improve."
- Carol Dweck

1. I can't do ______________________________

   ______________________________ ...YET.

2. ______________________________

   ____

   doesn't make sense … YET.

3. I don't know ______________________________

   ______________________________ ...YET.